

- 2 (a) A field of force is a region of space in which a body experiences a force without physical contact with another body. [The field is produced by the presence of another body (or bodies)]. [B1]
- (b) State one similarity and one difference between gravitational field and electric field.

Similarity [B1]:

- The field strength due to a point charge/mass follows an inverse square law with distance from the charge/mass.
- The potential due to a point charge/mass varies inversely with distance from the charge/mass.
- (Both G-fields and E-fields can be represented by field lines) In notes, we accept for students who memorised.

Difference [B1]:

- Gravitational fields are always attractive, whereas electric fields can be attractive or repulsive.
- Gravitational potential is always negative, whereas electric potential can be positive or negative.
- Source of gravitational field is a mass while source of electric field is a charge.

- (c) (i) At point P, it is a gravitational neutral point where net gravitational field strength g is zero. [B1]

It is closer to the moon since the mass of the moon is smaller than the mass of the Earth. [B1]

- (ii) At point P, *gravitational neutral point*.

$$g_M = g_E$$

$$G \frac{M_M}{r^2} = G \frac{M_E}{(3.80 \times 10^8 - r)^2} \quad [\text{M1- apply } g = G \frac{M}{r^2}]$$

$$\frac{(3.80 \times 10^8 - r)^2}{r^2} = \frac{5.98 \times 10^{24}}{7.35 \times 10^{22}} \quad [\text{B1}]$$

$$r = 3.79 \times 10^7 \text{ m} \quad [\text{A0}]$$

OR using potential method to solve,

Sum of potentials at point P (give method mark if approach is shown)



(iii) By conservation of energy

loss in KE = gain in GPE

$$\frac{1}{2}(m)(v)^2 - 0 = m(\phi_f - \phi_i)$$

$$\frac{1}{2}(v)^2 - 0 = (-1.3 - (-62.3)) \times 10^6 \quad [\text{C1}]$$

$$v = 11018$$

$$= 1.10 \times 10^4 \text{ m s}^{-1} \quad [\text{A1}]$$

The space probe needs to have sufficient energy from the Earth's surface to just reach the point P with maximum potential. Thereafter, the net gravitational force on the space probe (pointing towards Moon) will enable it to reach the Moon. [B1]

(d) (i) P is negative.

Near P, the electron has a positive potential energy. [M1]

Being very close to P, potential energy due to P on the electron is more significant than potential energy due to Q on the electron. (i.e.

$$\text{approximately } E_P \approx \frac{Q_P e}{4\pi\epsilon_0 r})$$

For potential energy to be positive, P and electron must be of the same sign. Hence P is negative. [A1]

OR

P is negative.

Potential energy increases towards P.[M1]

Since positive work must be done on electron to move it towards P.

Therefore, P and electron must be repelling each other.

Hence P is negative. [A1]

OR

P is negative.

Gradient of the graph is negative. [M1]

Since force points in the direction of decreasing potential energy ($F = -\frac{dU}{dr}$)

electrostatic force on electron must be in the direction of positive r (right).

Hence P and electron must be repelling each other

Hence P is negative. [A1]

(ii) Magnitude of force = gradient of potential energy graph [C1]

Gradient of graph (at x = 3.0 cm)

$$= \frac{2.3 - (-1.8)}{(0.0115 - 0.0445)} = 124 \text{ eV m}^{-1} \text{ [C1: gradient within range]}$$

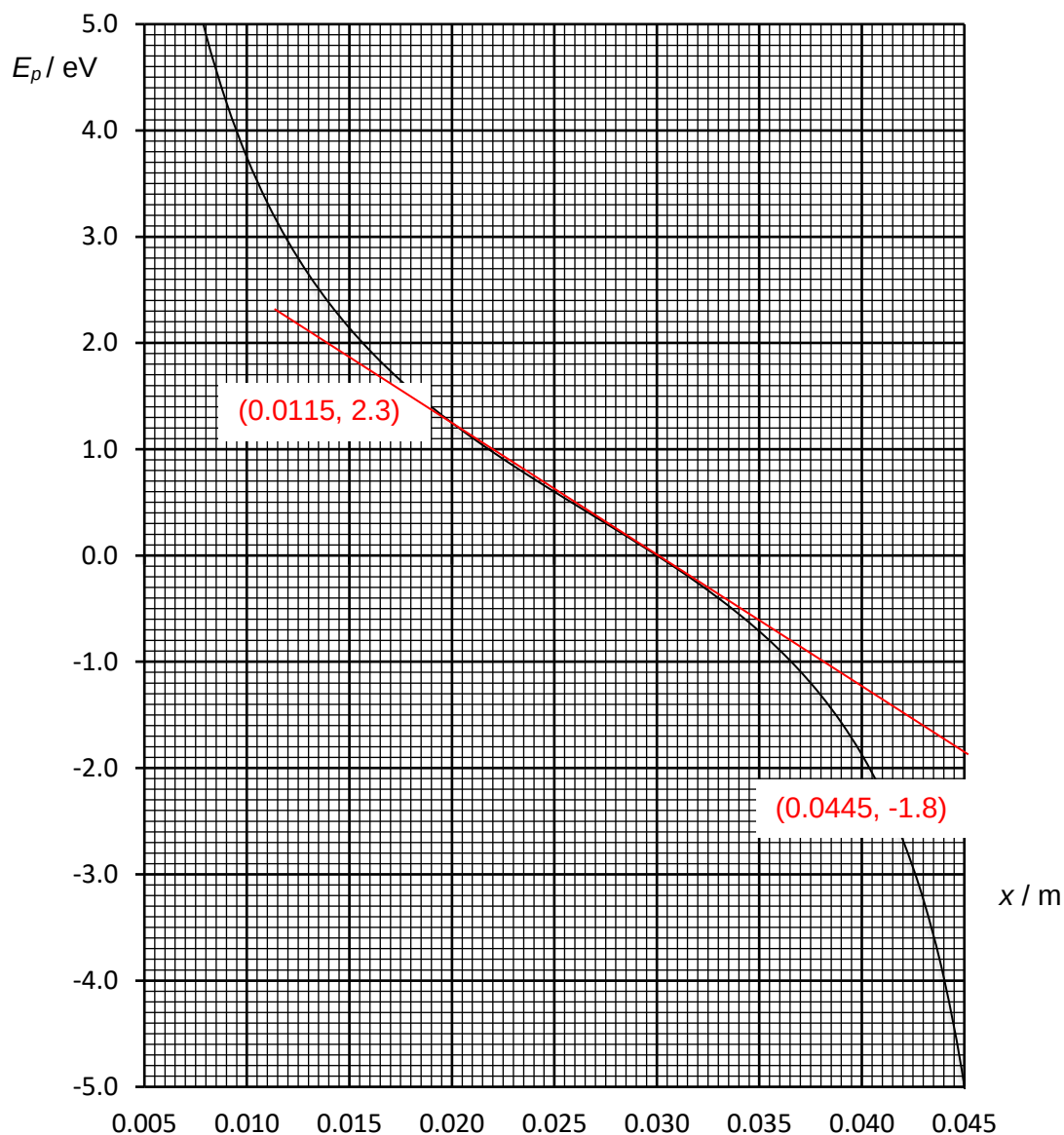
Accept gradient between 110 to 135 eV m^{-1}

Convert the unit to SI unit, force = $124 \times 1.6 \times 10^{-19} = 1.98 \times 10^{-17} \text{ N}$
 [A1:conversion to SI]

Accept a range of force due to gradient.

$$1.76 \times 10^{-17} \text{ N}$$

$$2.16 \times 10^{-17} \text{ N}$$



(iii) Curve with value of E_p near P to be lower & steeper gradient nearer to Q.

